

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTHM505 Project Management

Max Marks: 60

Date: 10/07/2024

Duration: 3.00 Hr.

/Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

A) Explain Phases in Project Management (CO1) 6

B) A project schedule has following characteristics (CO1) 6

Activity	A	B	C	D	E	F	G	H	I	J	K	L
	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time (Days)	4	1	1	1	6	5	4	8	1	2	5	7

Construct a Network Diagram

Compute the earliest event time and latest event time.

Determine the critical path and total project duration

C) Find the Total float and free float associated with the project in Q.1 (B) (CO1) 6

Q.2 Solve Any Two of the following.

A) Explain briefly Direct Cost and Indirect Cost for construction project. (CO2) 6

B) Consider the following table summarizing details for project.

Activity		A	B	C	D	E	F	G	H	I	J
Predecessors		-	-	-	A	B	C	C	E,F	D	H,G
Duration (week)	To	5	1	1	1	1	1	2	4	2	2
	T _m	6	3	4	2	2	5	2	4	5	2
	T _p	6	3	4	2	3	5	3	5	5	3

Construct the project network diagram.

Find the expected duration and variance of each activity.

Find the critical path and expected project completion of time. (CO3) 6

C) What is the probability of completing the project on or before 22 weeks? In Q.2. (B) (CO3) 6

Q.3 Solve Any Two of the following.

A) Write short notes on Central limit Theorem. (CO3) 6

B) Differentiate between resource leveling and resource smoothing. (CO2) 6

C) Explain steps in Resource Updating with an example. (CO2) 6

Q.4 Solve Any Two of the following.

- A) Write short notes on Demand and Supply. (CO4) 6
- B) A company has following details: (CO4) 6
- Fixed Cost = Rs. 20,00,000/-
- Variable Cost per Unit = Rs.100
- Selling price per Unit = Rs.200/-
- Find
- (a) The break even sales quantity
- (b) The break even sales
- C) Write short notes on Inflation. (CO4) 6

Q.5 Solve Any Two of the following.

- A) Explain use of Computers in Project Management. (CO5) 6
- B) Explain in detail the concept of Quality Control. (CO5) 6
- C) Explain Total Quality Management (TQM) with special reference to Philosophy of Juran. (CO5) 6

*** End ***